



Experimental and kinetic modeling study on 1,3-cyclopentadiene oxidation and pyrolysis

Hongyan Wang, Zhiqiang Liu, Siyuan Gong, Yujie Liu, Li Wang, Xiangwen Zhang, Guozhu Liu*

Key Laboratory for Green Chemical Technology of Ministry of Education, School of Chemical Engineering and Technology, Collaborative Innovation Center of Chemical Science and Engineering (Tianjin), Tianjin University, Tianjin 300072, China

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ABSTRACT

The oxidation of 1,3-cyclopentadiene ($c\text{-C}_5\text{H}_6$), a key intermediate formed in the combustion of aromatic and cyclic hydrocarbon fuels and formation of polycyclic aromatic hydrocarbons (PAHs), was studied in a jet-stirred reactor (JSR) under the pressure of 0.1 MPa and temperatures from 623 to 1073 K with three equivalence ratios ($\varphi = 0.5, 1.0$ and 1.8). The mole fraction profiles of 25 oxidation products including light hydrocarbons, oxygenated and aromatic species were identified and quantified using online synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) with uncertainties of C, H, and O molar balances about 24%, 20% and 14%, respectively. A detailed kinetic model with 245 species and 1638 reactions was constructed based on the theoretical calculation and reported models in literature and validated with the new JSR oxidation data in this work. The dominant consumption channels of 1,3-cyclopentadiene are through H-atom abstraction reactions by OH, HO_2 and CH_3 radicals to produce cyclopentadienyl radical ($c\text{-C}_5\text{H}_5$) under all equivalence ratios and the contribution of CH_3 attacking increases from $\varphi = 0.5$ to 1.8 . Subsequently, the $c\text{-C}_5\text{H}_5$ is consumed via the reactions with HO_2 and O_2 to form small molecule and the recombination reaction with $c\text{-C}_5\text{H}_6$ to yield PAHs. Furthermore, the current model was also validated with a wide range of experimental data in literature of 1,3-cyclopentadiene combustion, including the species concentration profiles in flow reactors pyrolysis and oxidation.

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1. Introduction

1,3-Cyclopentadiene ($c\text{-C}_5\text{H}_6$) is a significant component with high concentration levels in the pyrolysis of many high-energy density fuels and biomass fuels such as 2,5-dimethylfuran (DMF) [1], quadricyclane (QC) [2–5], tetrahydrotricyclopentadiene (THTCPD) [6], exo-tetrahydrodicyclopentadiene (exo-TCDD) [7,8], etc. Meanwhile, due to its weak C–H bond at 5 position, 1,3-cyclopentadiene is easily converted into the cyclopentadienyl radical ($c\text{-C}_5\text{H}_5$) which is also recognized as a significant transition product between aromatic and aliphatic during the oxidation of mono-alkylated aromatic hydrocarbons [9–15]. For instance, $c\text{-C}_5\text{H}_5$ is abundantly produced through decarbonylation of phenoxy radical mainly from the combination of phenyl and oxygen during the oxidation of benzene [11]. Furthermore, the combustion chemistry of $c\text{-C}_5\text{H}_6/c\text{-C}_5\text{H}_5$ system also plays a significant role in the formation of polycyclic aromatic hydrocarbons (PAHs) [16–18], which are important combustion-derived intermediates known

to be the vital soot precursors as well as inherent mutagenic and carcinogenic characteristics [19]. The recent researches show that PAHs can be produced via the recombination reaction of $c\text{-C}_5\text{H}_5$ with $c\text{-C}_5\text{H}_6$ and chain addition reactions of $c\text{-C}_5\text{H}_5$ with acetylene or vinyl radical [18]. Therefore, the experimental and kinetic modeling study on the 1,3-cyclopentadiene combustion chemistry is crucial to improve the understanding of the pyrolysis and oxidation processes of cyclic hydrocarbons.

The most previous experimental studies were mainly focused on the thermal decomposition of $c\text{-C}_5\text{H}_6$ and emphasized the importance of $c\text{-C}_5\text{H}_6/c\text{-C}_5\text{H}_5$ in the PAHs formation using the shock tubes [20,21] and flow reactors [16,22–25]. By contrast, few experimental researches have been conducted on the $c\text{-C}_5\text{H}_6$ oxidation. In 2001, Burcat et al. [26] measured the ignition delay times of $c\text{-C}_5\text{H}_6$ (diluted in Ar) in a single pulse shock tube over the temperature range of 1278–2110 K and pressures of 0.25–1.25 MPa with equivalence ratios from 0.5 to 2.0. Subsequently, Orme et al. [27] also studied the ignition delay times of $c\text{-C}_5\text{H}_6$ in a shock tube under atmospheric pressure, temperatures of 1300–2200 K and the equivalence ratios of 0.5–2.0. Ji et al. [28] investigated the laminar flame speeds and extinction strain rates

* Corresponding author.

E-mail address: gliu@tju.edu.cn (G. Liu).

of $c\text{-C}_5\text{H}_6$ /air mixture in the counter-flow configuration at the pressure of 0.1 MPa, unburned mixture temperature of 353 K, and a wide range of equivalence ratios ($\varphi=0.7\text{--}1.5$). They simulated the experimental results using two available kinetic models in previous literature (USC Mech II model [29] and toluene model of Dooley et al. [30]) and proposed the uncertainties of the $c\text{-C}_5\text{H}_5$ consumption pathways including $c\text{-C}_5\text{H}_5 \rightarrow \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$ and $c\text{-C}_5\text{H}_5 + \text{OH} \rightarrow (1,3\text{-butadiene}) \text{C}_4\text{H}_6 + \text{CO}$. Compared to these macroscopic combustion parameters studies, the investigation of detailed species distributions in the 1,3-cyclopentadiene oxidation are relatively limited. Butler and Glassman [23] studied the $c\text{-C}_5\text{H}_6$ oxidation (diluted in N_2) in Princeton's adiabatic flow reactor at the pressure of 0.1 MPa, initial temperatures of 1100–1200 K and equivalence ratios of 0.6–1.6. They obtained the speciation profiles of $\text{C}_1\text{--C}_{14}$ using off-line GC-FID and suggested that the cyclopentadienoxo radical ($c\text{-C}_5\text{H}_5\text{O}$) was an important intermediate, which can convert into $n\text{-butadienyl} + \text{CO}$ or 2,4-pentadienal-5-yl via the ring-opening reaction. Though previous experimental work provided a basic understanding of 1,3-cyclopentadiene oxidation, it still needs further investigation on the consumption paths of $c\text{-C}_5\text{H}_5$ to produce small species (e.g. C_3H_3 , C_2H_2 or C_4H_6) or aromatic products. In addition, it should be noted that the existing experimental data mainly concentrated on the relatively high temperature region ($>1000\text{ K}$) and the kinetic models developed to predict the 1,3-cyclopentadiene combustion chemistry cannot be extensively validated. More importantly, the study of the low-temperature oxidation of 1,3-cyclopentadiene is even absent in the measurement of microscopic parameters concerning the reactive intermediates and stable products profiles. The low-temperature kinetic mechanism plays an extremely essential role in revealing the ignition process and verifying the universality of the developed models. Hence, the low-temperature oxidation study of 1,3-cyclopentadiene is meaningful to give more information on the reaction pathways and improve the understanding of $c\text{-C}_5\text{H}_6$ chemistry.

There are also a large amounts of theoretical researches [31–54] on the reactions involving $c\text{-C}_5\text{H}_6$ and $c\text{-C}_5\text{H}_5$, which will be described in the Development of the kinetic model Section 3. Although recent kinetic models usually consider the reaction pathways of $c\text{-C}_5\text{H}_6$ to produce aromatic species, most of these were neither exclusively constructed nor comprehensively verified for 1,3-cyclopentadiene chemistry. For example, Yuan et al. [13] proposed a detailed kinetic mechanism of toluene pyrolysis and oxidation with 272 species and 1698 reactions involving $c\text{-C}_5\text{H}_6$ and predicted the corresponding pyrolysis data in literature [12]. Djokic et al. [16] refined the mechanism for $c\text{-C}_5\text{H}_6$ pyrolysis based on the POLIML_1311 model [36] and successfully simulated their pyrolysis experimental data. Zhou et al. [55] updated the mechanism containing $c\text{-C}_5\text{H}_6$ chemistry on the basis of Aramco 2.0 model that was validated against a large array of experimental measurements of $\text{CO}\text{--C}_4$ species. Although these model all contain $c\text{-C}_5\text{H}_6$ mechanism and are validated with some pyrolysis data, the more comprehensive comparison with $c\text{-C}_5\text{H}_6$ oxidation experimental data is still lacking.

The objective of this work is to investigate the low temperature oxidation of $c\text{-C}_5\text{H}_6$ in a jet stirred reactor at the pressure of 0.1 MPa, temperatures of 623–1073 K and equivalence ratios of 0.5, 1.0 and 1.8. The online synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) was used to identify reactive intermediates and stable products of $c\text{-C}_5\text{H}_6$ oxidation and measure their mole fraction profiles as the function of reaction temperature. On the basis of recent theoretical progresses and reported models in literature, a comprehensive kinetic model of $c\text{-C}_5\text{H}_6$ was constructed and validated against present and literature experimental data covering various reaction circumstances. The rate of production (ROP) and sensitivity analysis were

performed to provide a deeper understanding of $c\text{-C}_5\text{H}_6$ oxidation and aromatic growth.

2. Experiments

2.1. Materials

The 1,3-cyclopentadiene used in the current work was obtained by distilling dicyclopentadiene to depolymerize in a flask at 443–453 K and collecting distillates at 314–317 K. The purity of 1,3-cyclopentadiene is about 99.3%, shown in Fig. S1 of the Supplementary data 1. The dicyclopentadiene ($>97\%$) was purchased from J&K Scientific Corporation Limited, Beijing. The oxygen and argon were provided by Nanjing Gas Company, China with purities of 99.99% and 99.999%, respectively.

2.2. Experimental method

The low-temperature oxidation experiments of $c\text{-C}_5\text{H}_6$ were carried out in a jet stirred reactor (JSR) at National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The detailed information of the BL03U beamline and apparatus used in this work have been described in earlier publications [56,57], thus only a brief introduction is given here. The experimental equipment is consisted of three parts, including a jet stirred reactor, a differentially pumped chamber with a molecular beam sampling system, and a photoionization chamber with a home-made reflectron time-of-flight (TOF) mass spectrometer (MS). The JSR is composed of a small fused silica sphere (I.D.: 58.0 mm and T: 2.0 mm) and four turbulent jets located at the center of the sphere, which can provide stirring and ensure the mixing of the gas mixture to achieve homogeneity in composition and temperature. The detailed verification of homogeneity in Hefei JSR is given in the Supplementary data 1. A quartz annular preheating zone attached to the reactor is used to preheat the gas mixture with a very short time (only a few percent of the residence time) to improve the temperature homogeneity in the reactor [58]. The heating of the reactor and preheating zones is in an electrical oven and the reaction temperature is measured using a K-type thermocouple located at the center of the reactor.

During the experiments, liquid $c\text{-C}_5\text{H}_6$ with the flow rate of 0.037 ml/min (1.0% in gas phase) was heated to 373 K above boiling point (314 K) in a vaporizing tank. Subsequently, it was mixed with O_2 and Ar in a 3.0 m long stainless-steel tube into JSR, whose flow rates were controlled separately by mass flow controllers. All transfer tubes were maintained at 473 K using heating belts before entering JSR to minimize temperature gradients and after flowing out of JSR to avoid condensation. Table S1 of Supplementary data 1 lists the detailed experimental conditions of JSR oxidation of $c\text{-C}_5\text{H}_6$ in the current work.

During the experimental processes, two kinds of experimental modes were adopted. At a fixed temperature (848 K) and equivalent ratio (0.5), the photoionization efficiency (PIE) spectra (7.8–10.3 eV) scans were performed to give the information of ionization energies to identify $c\text{-C}_5\text{H}_6$ oxidation products. Over a variety of temperatures, the mass spectra were obtained at the photon energies of 16.64, 12.0, 11.0, 10.0, 9.5, 9.0 and 8.5 eV to obtain mole fraction profiles of oxidation species. Additionally, the cold gas experiments were conducted at 16.64 eV for H_2 , CH_4 , C_2H_2 , O_2 , CO and CO_2 to measure the values of photoionization cross section and mass discrimination factor. The methods of products identification and mole fraction evaluation have been previously given in detail [56]. The quantification of species requires the information of the photoionization cross sections (PICS) which are available in the online database [59]. The total mass balance for measured species is within 3% uncertainty in comparison to carbon, hydrogen and oxygen inputs at all experimental conditions (Fig. 2 of the

Supplementary data 1), which can indicate the reasonability the evaluation of mole fractions. The uncertainties of C, H, and O molar balances are 24%, 20% and 14%, respectively, which may come from the uncertainty of the photoionization cross-section for some species and the undetected formation large oxygenated products. More experimental details are given in the Supplementary data 1.

3. Development of the kinetic model

A detailed kinetic model of c-C₅H₆ oxidation with 245 species and 1638 reactions was constructed based on the theoretical calculation and reported models in literature, including the sub-mechanisms of C0–C4 species, 1,3-cyclopentadiene and aromatic hydrocarbons (C6–C10). The C0–C4 species were mainly taken from the USC Mech II model [29] and partly from the Aramco Mech 3.0 model [60] and the model of acetylene low temperature oxidation [61], which were validated against the comprehensive combustion scenarios. The detailed construction processes of c-C₅H₆ and C6–C10 aromatic species sub-mechanisms are described below. In Supplementary data 1, Table S2 provides the rate constants and sources of partial important reactions in the current model and Table S3 gives the C5 and larger species information including formulas, names, molecular weights, nomenclatures and structures.

3.1. 1,3-Cyclopentadiene sub-mechanism

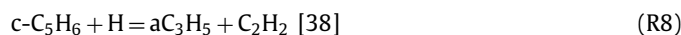
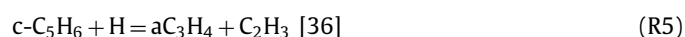
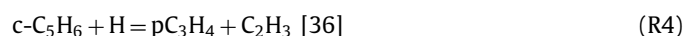
3.1.1. Unimolecular decomposition of 1,3-cyclopentadiene

The unimolecular decomposition of c-C₅H₆ can generate c-C₅H₅ and allene (aC₃H₄)+acetylene (C₂H₂) through R1 and R2, respectively. The rate constant of R1 was widely investigated by many groups using theoretical and experimental methods [16,20,21,32,34–36,46,53,62], shown in Fig. S2(a)) of Supplementary data 1. Among them, Colket [53] first measured the rate coefficients of R1 in a shock tube at temperatures of 1100–2000 K and pressures of 1–1.3 MPa. Their rate constants were generally located in the middle of the literature value distributions and adopted in the current model. The rate constant of R2 was taken from the theoretical work of Bacskay and Mackie [35] using the CASSCF/CASPT2 method and RRKM theory.

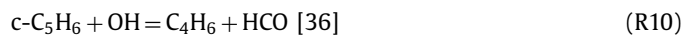
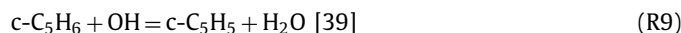


3.1.2. Bimolecular decomposition and oxidation of 1,3-cyclopentadiene

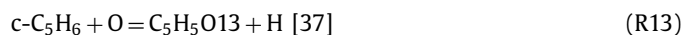
The H-abstraction reaction of c-C₅H₆ by H can form c-C₅H₅ (R3) and its rate constant was taken from the model of Burcat and Dvinyaninov [20]. The combination reactions between c-C₅H₆ and H can produce propyne (pC₃H₄)+vinyl radical (C₂H₃) (R4), aC₃H₄+C₂H₃ (R5) and propargyl radical (C₃H₃)+ethylene (C₂H₄) (R6), whose the rate constants were taken from the complete POLIMI_1311 model [36]. Moreover, Zhong and Bozzelli [37] proposed that the addition reaction of H atom at the 1 position of c-C₅H₆ can produce a linear 1,4-pentadien-3-yl radical (IC₅H₇) by opening the 5- carbon ring (R7) and calculated the rate constants of R7 and R8 (allyl radical (aC₃H₅)+C₂H₂), which were adopted in this model. The subsequent reaction of IC₅H₇ leading to aC₃H₅+C₂H₂ was taken from the USC Mech II model [29]. The H-atom abstraction reaction of c-C₅H₆ by CH₃ to form c-C₅H₅ was taken from the model of Vervust et al. [38].



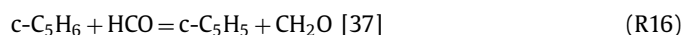
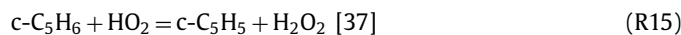
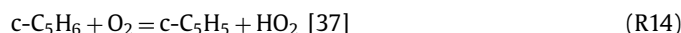
The reactions between c-C₅H₆ and OH involve the H abstraction reaction to produce c-C₅H₅ (R9) and the open-ring reaction to produce C₄H₆+HCO (R10). The rate constants of R9 and R10 were derived from the recommended value of Emdee et al. [39] and POLIMI_1311 model [36], respectively.



Similar to the reactions between c-C₅H₆ and OH, the reactions of c-C₅H₆ with O also contain the H abstraction reaction to produce c-C₅H₅ (R11) and the open-ring reaction to produce C₄H₆+CO (R12), which were taken from the POLIMI_1311 model [36]. 1-oxy-1,3-cyclopentadiene radical (C₅H₅O13) can be formed through replacing an H atom with O of c-C₅H₆ (R13) and the rate constant was taken from the calculation of Zhong and Bozzelli [37] in the current model.

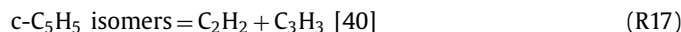


In addition, the H abstraction reactions of c-C₅H₆ by O₂, HO₂ and HCO can produce c-C₅H₅ with HO₂, H₂O₂ and formaldehyde (CH₂O) (R14–R16), respectively. The rate constants calculated by Zhong and Bozzelli [37] were used for R14–R16 in the current model.



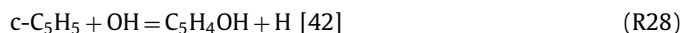
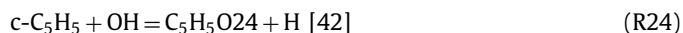
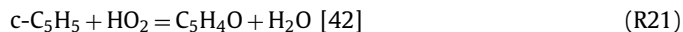
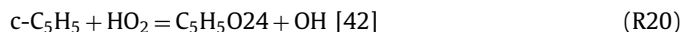
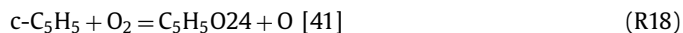
3.1.3. Decomposition and oxidation of 1,3-cyclopentadienyl radical

As proposed by Moskaleva and Lin [40], the unimolecular isomerization of c-C₅H₅ can produce a cyclic isomer (C₅H₅2) through the 1,2-H transfer reaction and the 3-pentene-1-yne-5-yl radical (IC₅H₅) via the C–C bond dissociation reaction. Subsequently, the unimolecular decomposition of c-C₅H₅ and its isomers can generate C₂H₂+C₃H₃ (R17). The relative pressure-dependent rate constants of these reactions were calculated by Moskaleva and Lin [40] and widely used in models involving 1,3-cyclopentadienyl.



The oxidation of c-C₅H₅ occurs mainly through association reactions with O₂, HO₂, O and OH. The reactions between c-C₅H₅ and O₂ can produce 1-oxy-2,4-cyclopentadiene radical (C₅H₅O24) (R18) and 2,4-cyclopentadiene-1-one (C₅H₄O) (R19), which were similar to the reactions of c-C₅H₅ with HO₂ (R20–21), O (R22–23) and OH (R24). The reaction of c-C₅H₅ with O can also produce butadiene radical (nC₄H₅)+CO (R25) and the reaction of c-C₅H₅ with OH can form C₄H₆+CO (R26), 1-OH-2,4-cyclopentadien-1-yl radical (C₅H₅OH) (R27) and 2,4-cyclopentadien-1-ol (C₅H₄OH)+H (R28). Murakami et al. [41] measured the rate constant of R18 in a reflected shock, which was incorporated in the current model.

Zhong and Bozzelli [42] studied the high-pressure limit rate constants of those reactions (R19–R28), which were widely used in previous models and also adopted in the current model.



3.1.4. Decomposition and oxidation of C5 ring oxygenates

Based on the decomposition and oxidation reactions of $\text{c-C}_5\text{H}_6$ and $\text{c-C}_5\text{H}_5$, five C5 ring oxygenates are produced, including $\text{C}_5\text{H}_5\text{OH}$, $\text{C}_5\text{H}_4\text{OH}$, $\text{C}_5\text{H}_5\text{O}_2$, $\text{C}_5\text{H}_5\text{O}_3$, and $\text{C}_5\text{H}_4\text{O}$. The unimolecular decomposition of $\text{C}_5\text{H}_5\text{OH}$ can produce $\text{C}_5\text{H}_4\text{OH}$ and then $\text{C}_5\text{H}_4\text{OH}$ can form $\text{C}_5\text{H}_4\text{O}$ by the C–H bond cleavage reaction, which also can be generated by the unimolecular decomposition reactions of $\text{C}_5\text{H}_5\text{O}_2$ and $\text{C}_5\text{H}_5\text{O}_3$. And the dissociation reactions of $\text{C}_5\text{H}_5\text{O}_2$ and $\text{C}_5\text{H}_4\text{O}$ can produce nC_4H_5 . These reactions were incorporated in the current model and rate constants were referred to theoretical calculations of Robinson and Lindstedt [43]. In addition, the reactions between $\text{C}_5\text{H}_5\text{OH}$ and H can produce $\text{C}_5\text{H}_5\text{O}_2$ and $\text{C}_5\text{H}_4\text{OH}$ and the rate constants of both them were derived from the theoretical results by Zhong and Bozzelli [42].

The bimolecular H-assisted reaction of $\text{C}_5\text{H}_4\text{OH}$ can convert into $\text{C}_5\text{H}_4\text{O}$ and subsequent the H-abstraction reaction of $\text{C}_5\text{H}_4\text{O}$ can produce 5-one-1,3-cyclopentadien-1-yl ($\text{C}_5\text{H}_3\text{O}$) + H_2 . The $\text{C}_5\text{H}_3\text{O}$ can lose the CO group to transform $\text{C}_2\text{H}_2 + \text{C}_2\text{H}$ by the ring opening reaction. The oxidation reactions of $\text{C}_5\text{H}_5\text{OH}$, the reactions of $\text{C}_5\text{H}_5\text{OH}$ with O, OH and HO_2 are consistent and can form $\text{C}_5\text{H}_4\text{OH}$ and $\text{C}_5\text{H}_5\text{O}_2$. These reactions were considered in the current model and rate constants were taken from the model of Alzueta et al. [63], which was validated with the benzene oxidation in a flow reactor at 0.1 MPa and 900–1450 K.

3.2. Aromatic hydrocarbons sub-mechanism

3.2.1. Formation and consumption of monocyclic aromatic hydrocarbons (MAHs)

Benzene (A1) is the simplest of MAHs with molar mass larger than fuel in the $\text{c-C}_5\text{H}_6$ oxidation and three major paths of A1 formation are considered in the current model. Firstly, A1 can be produced from the $\text{C}_{10}\text{H}_{11}$ adduct via a tricyclic intermediate, in which the $\text{C}_{10}\text{H}_{11}$ adduct was formed by the addition reaction of $\text{c-C}_5\text{H}_6$ and $\text{c-C}_5\text{H}_5$. Cavallotti et al. [64] identified the potential energy surface (PES) and calculated the global kinetic constant of $\text{c-C}_5\text{H}_6 + \text{c-C}_5\text{H}_5 \rightarrow \text{A1} + \text{nC}_4\text{H}_5$, which was successfully applied to many models [13,65–69] and thus adopted in the current model. Secondly, the pathways of $\text{C}_3 + \text{C}_3$ and $\text{C}_4 + \text{C}_2$ formed A1 [60,70,71] were considered in the current model. Lastly, A1

was also suggested to be produced from the reaction between CH_3 and $\text{c-C}_5\text{H}_5$ [17,45,72]. Melius and Colvin [17] and Moskaleva et al. [72] calculated the PES of $\text{c-C}_5\text{H}_5 + \text{CH}_3$ and proposed the recombination can produce 5-methyl-cyclopentadiene ($\text{C}_5\text{H}_5\text{CH}_3$ -5), which can isomerize into 1-methyl-cyclopentadiene ($\text{C}_5\text{H}_5\text{CH}_3$ -1) and 2-methyl-cyclopentadiene ($\text{C}_5\text{H}_5\text{CH}_3$ -2). These three species were lumped as $\text{C}_5\text{H}_5\text{CH}_3$ due to the similar energies in the current model. The $\text{C}_5\text{H}_5\text{CH}_3$ can convert into fulvene ($\text{C}_5\text{H}_4\text{CH}_2$) and H_2 , and then the H atom assisted reaction of $\text{C}_5\text{H}_4\text{CH}_2$ can produce A1. This process was re-elucidated by Sharma and Green [45] using more precise theoretical methods and pressure-dependent rate constants were calculated by RRKM/master equation. Their rate constants were very close to the experimental measurements of Ikeda et al. [46] and Lifshitz et al. [47] and were adopted in the current model, shown in Fig. S2(b)) of Supplementary data 1.

The consumption paths of A1 are mainly via the unimolecular decomposition and H-atom abstraction reactions by H and CH_3 radicals to produce phenyl radical (A1-), which can further decompose to form benzyne ($\text{o-C}_6\text{H}_4$). The decomposition of $\text{o-C}_6\text{H}_4$ can produce diacetylene (C_4H_2) and C_2H_2 . Additionally, the oxidation of A1- can generate phenoxyl radical (A1O), which can produce $\text{c-C}_5\text{H}_5 + \text{CO}$ through the unimolecular decomposition reaction. The rate constants of these reactions were taken from the benzene models available in literature [10,12,63,68].

Two main formation routes for toluene (A1CH_3) were considered in the current model, including $\text{A1} + \text{CH}_3$ and $\text{c-C}_5\text{H}_6 + \text{C}_2\text{H}_3$. The rate constants of both reactions were taken from the model of Vervust et al. [38]. Ethylbenzene ($\text{A1C}_2\text{H}_5$) can be formed by CH_3 addition to benzyl (A1CH_2), which can be yield through unimolecular decomposition and H-abstraction reactions of A1CH_3 by OH, H, CH_3 , O, etc. Sharma et al. [73] proposed the recombination reaction of $\text{c-C}_5\text{H}_5 + \text{C}_3\text{H}_3$ to produce styrene ($\text{A1C}_2\text{H}_3$) using the CBS-QB3 method. The $\text{A1C}_2\text{H}_3$ can also be generated from the stepwise H elimination reactions of $\text{A1C}_2\text{H}_5$. The phenylacetylene ($\text{A1C}_2\text{H}$) can be formed from the combination reaction of $\text{A1-} + \text{C}_2\text{H}_2$ and H elimination reactions of $\text{A1C}_2\text{H}_3$. The rate constants of these four MAHs were mainly taken from existing models in literature [13,74,75].

3.2.2. Formation and consumption of polycyclic aromatic hydrocarbons (PAHs)

The important bicyclic PAHs include indene (C_9H_8) and naphthalene (A2), which are abundantly formed in the pyrolysis and oxidation of $\text{c-C}_5\text{H}_6$. An important pathway to produce C_9H_8 through the addition of $\text{c-C}_5\text{H}_5$ to $\text{c-C}_5\text{H}_6$ has been widely investigated in literature [38,49,64,76,77]. Recently, Vervust et al. [38] updated the high-pressure limit rate constants of the route to C_9H_8 , which were tested by the simulations of $\text{c-C}_5\text{H}_6$ pyrolysis experiments and adopted in the current model. The combination reaction between $\text{c-C}_5\text{H}_5$ and C_2H_2 can produce the heptatrienyl radical (cC_7H_7) and then cC_7H_7 can add another C_2H_2 to form C_9H_8 . The rate constants of them were derived from the theoretical calculation of Fascella et al. [54]. In addition, the reactions of $\text{A1CH}_2 + \text{C}_2\text{H}_2$ and $\text{A1-} + \text{aC}_3\text{H}_4$ can also form C_9H_8 and the rate constants were taken from the results of Vereecken et al. [78,79]. The consumption pathways of C_9H_8 mainly include the H abstraction reactions by H, CH_3 , O, OH and HCO to produce indenyl radical (C_9H_7) and the rate constants were taken from the soot formation model of Blanquart et al. [80]. The rate constants of C_9H_7 decomposition reactions were taken from the analogous results by Yuan et al. [13].

The naphthalene (A2) can be produced from the self-recombination reaction of $\text{c-C}_5\text{H}_5$, which has attracted a lot of theoretical and experimental researches. Some previous kinetic models [49–51,64,81,82] considered this pathway as a simplified single-step global reaction ($\text{c-C}_5\text{H}_5 + \text{c-C}_5\text{H}_5 = \text{A2} + 2\text{H}$) and proposed high pressure limit rate coefficients, which was not an

adequate description for A2 formation process. More recently, Long et al. [52] re-calculated the PES of $c\text{-C}_5\text{H}_5$ self-recombination to A2. The results involved fulvalanyl radical isomers (N1a, N1b and N1c) and the azulanyl radical isomers (N7a, N7b and N7c) pathways, which were initiated by the self-recombination of $c\text{-C}_5\text{H}_5$ to produce six fulvalane isomers (T1a to T1f). The fulvalane isomers can either isomerize to azulane isomers (T4a to T4h) or directly decompose to fulvalanyl isomers + H. The fulvalanyl isomers can either directly decompose to generate fulvalene (FlvIn) or isomerize to the dihydronaphtyl radical, which can lose H to produce A2. Similarly, the azulanyl isomers can either generate azulene (AzIn) + H or isomerize to the dihydronaphtyl radical to produce A2. Long et al. [52] determined the pressure-dependent rate coefficients of the process, which were successfully used in the copyrolysis model of $c\text{-C}_5\text{H}_6$ and C_2H_4 of Vervust et al. [38]. Thus, the pressure-dependent rate constants of these reactions were taken from the theoretical results of Long et al. [52] in the current model.

Moreover, A2 can also be produced by the recombination between A1CH_2 and C_3H_3 . Matsugi and Miyoshi [83,84] theoretically investigated the PES of $\text{A1CH}_2 + \text{C}_3\text{H}_3$ and suggested a multistep kinetic mechanism. The reaction of $\text{A1CH}_2 + \text{C}_3\text{H}_3$ can produce methyleneindanyl ($\text{C}_9\text{H}_7\text{CH}_2$) radical and then the $\text{C}_9\text{H}_7\text{CH}_2$ converts into methyleneindene ($\text{C}_9\text{H}_6\text{CH}_2$) or A2. The addition reaction of H to $\text{C}_9\text{H}_6\text{CH}_2$ can also generate A2. The relevant rate constants were derived from the study of Matsugi and Miyoshi [83,84]. The reaction of $\text{C}_9\text{H}_7 + \text{CH}_3$ can produce $\text{C}_9\text{H}_7\text{CH}_2$ via losing H atom and the rate constant was obtained by analogy with the reaction of $c\text{-C}_5\text{H}_5 + \text{CH}_3$ [45]. In addition, A2 was suggested to be formed from the reactions of fulvenallenyl radical (C_7H_5) + C_3H_3 and $\text{A1} + \text{C}_4$ species, in which C_7H_5 can be produced from the stepwise H-atom elimination reactions of A1CH_2 . The corresponding rate constants were derived from the theoretical work of Silva and Bozzelli [85] and Blanquart et al. [80].

The consumption pathways of A2 mainly include the H-atom abstraction reactions to form naphthyl radical (A2-) and the oxidation reactions to produce naphthalenoxyl radical (A2O), which A2O ultimately decomposes to C_9H_7 and CO. These reactions were considered in the current model and rate constants were taken from available models in literature [13,86].

3.3. Thermodynamic data and transport data

In the current model, most of the thermodynamic data and transport data of CO–C5 species were taken from the USC Mech II model [29] and AramcoMech 3.0 mechanism [60] and those of C6–C8 species were taken from the previous toluene models [13,36,82]. The source of thermodynamic data of those species, related with the formation of bicyclic PAHs, was consistent with that of reaction mechanism. The reaction mechanism, thermodynamic data and transport data are given in the Supplemental materials. In this work, the simulation work of different kinds of experiments were executed using respective physical models with the Chemkin-Pro software [87].

4. Results and discussion

In the current work, a total of 25 major species and significant intermediates including light hydrocarbons, oxygenated and aromatic species were detected and quantified using SVUV-PIMS during the $c\text{-C}_5\text{H}_6$ oxidation. A detailed kinetic model with 245 species and 1638 reactions was established to better understand $c\text{-C}_5\text{H}_6$ oxidation using a wide range of further validation. The previous pyrolysis experimental data by Butler et al. [23], Djokic et al. [16], Vervust et al. [38] and Kim et al. [22] and the oxidation experimental data by Butler et al. [23] in three kinds of flow

reactors were compared with the simulation obtained using the current model.

4.1. Jet-stirred reactor oxidation of 1,3-cyclopentadiene

The JSR oxidation of $c\text{-C}_5\text{H}_6$ was conducted at 0.1 MPa with three equivalence ratios (0.5, 1.0 and 1.8). The detailed experimental conditions are given in Table S1 of Supplementary data 1. The simulation work was carried out using the Perfectly Stirred Reactor model through the Chemkin-Pro software [87]. The experimental and simulated results are shown in Figs. 2–6. Generally the present model can reasonably reproduce the maximum mole fractions and initial formation temperatures for many products. The rate of production (ROP) analysis and sensitivity analysis were performed to provide insights into the $c\text{-C}_5\text{H}_6$ oxidation chemistry at different equivalence ratios. For $\varphi = 0.5$, 1.0 and 1.8, the temperatures of 848, 873 and 898 K were selected to perform these analyses, respectively, which correspond to around 50% conversion of $c\text{-C}_5\text{H}_6$. The reaction networks at $\varphi = 0.5$ and 1.8 are shown in Fig. 7 as an example.

4.1.1. Identification of oxidation products

Fig. S3 of Supplementary data 1 illustrates the lumped photoionization mass spectrum of $c\text{-C}_5\text{H}_6$ oxidation at 873 K and photon energies from 8.5 to 12.0 eV. With the increase of photon energy, the oxidation products with different ionization energies are ionized to produce more the mass spectrum peaks covering the CO–C10 species. A total of 25 major species were detected including Ar, H_2 , CH_4 , H_2O , C_2H_2 , CO, C_2H_4 , CH_2O , O_2 , CO_2 , CH_3CHO , aC_3H_4 , pC_3H_4 , propylene (C_3H_6), C_4H_6 , acrolein ($\text{C}_2\text{H}_3\text{CHO}$), $c\text{-C}_5\text{H}_5$, $c\text{-C}_5\text{H}_6$, $\text{C}_5\text{H}_4\text{CH}_2$, A1, A1CH_3 , $\text{A1C}_2\text{H}$, $\text{A1C}_2\text{H}_3$, C_9H_8 and A2.

It is worth mentioning that the reactive radical and isomers including $c\text{-C}_5\text{H}_5$, aC_3H_4 , pC_3H_4 , $\text{C}_5\text{H}_4\text{CH}_2$ and A1 were clearly identified. Fig. 1 shows the measured photoionization efficiency (PIE) spectra of $c\text{-C}_5\text{H}_6$, $c\text{-C}_5\text{H}_5$, aC_3H_4 , pC_3H_4 , $\text{C}_5\text{H}_4\text{CH}_2$ and A1. The PIE spectra are yielded and plotted versus the photon energy by integrating the ion signals of each mass peak normalized with photon flux [88], which are used to identify species by comparing ionization thresholds and molecular weight with literature. Fig. 1(a) shows the measured ionization threshold (8.58 eV) and the signal profile are in good agreement with the previous $c\text{-C}_5\text{H}_6$ measurement [89] (IE = 8.57 eV), indicating the signal of $m/z = 66$ measured in this work can be attributed to $c\text{-C}_5\text{H}_6$. Fig. 1(b) shows the PIE spectrum of $m/z = 65$ with an onset at 8.40 eV, which is related to the $c\text{-C}_5\text{H}_5$ ionization of the theoretical and experimental values [90,91] (IE = 8.41 eV). Fig. 1(c) shows the measured ionization thresholds of $m/z = 40$ are around 9.71 and 10.33 eV, which agree well with the measurements of aC_3H_4 [92] (IE = 9.69 eV) and pC_3H_4 [93] (IE = 10.36 eV), respectively. Analogously, Fig. 1(d) shows the measured ionization thresholds of $m/z = 78$ are 8.41 and 9.21 eV, corresponding to experimental measurements of $\text{C}_5\text{H}_4\text{CH}_2$ [94] (IE = 8.36 eV) and A1 [93] (IE = 9.24 eV).

4.1.2. Major species

The photoionization mass spectra measured at varied reactor temperatures were applied to obtain the mole fraction profiles of observed oxidation species. Figure S4 of Supplementary data 1 provides the photoionization mass spectra, which were measured with the photon energy of 12.0 eV at the reactor temperature of 811, 873, 978, 1006, 1026 and 1069 K, respectively. It can be seen that the mass spectrometry signal $m/z = 66$ corresponding to $c\text{-C}_5\text{H}_6$ parent molecule is gradually weakening with the increasing of temperature, while the type and mass spectrometry signal of oxidation products are increasing. Meanwhile, there are also some product signals that decompose and decline when they reach the peaks.

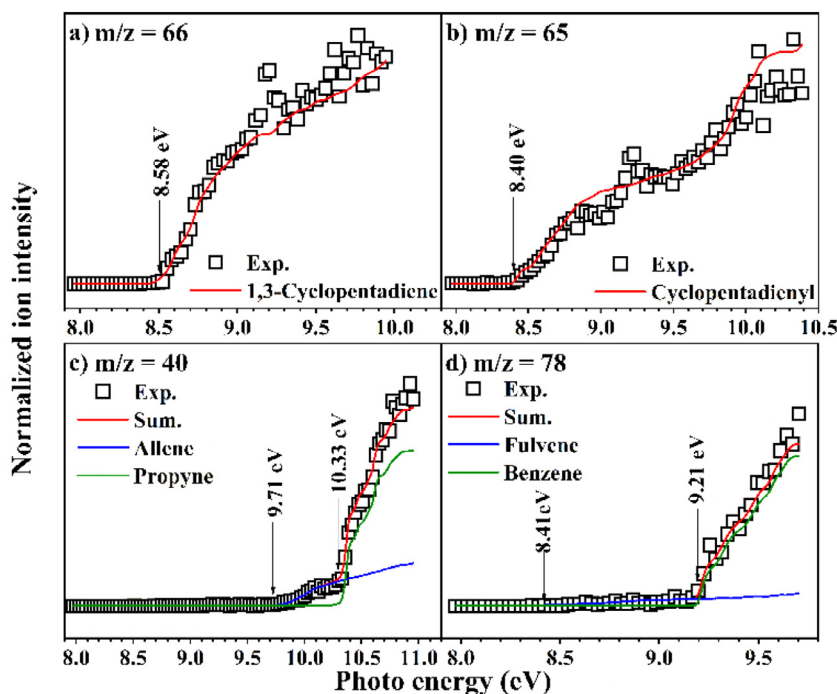


Fig. 1. The PIE curves vs. photo energy E for (a) $m/z = 66$, (b) $m/z = 65$, (c) $m/z = 40$, (d) $m/z = 78$ in the JSR oxidation of $c\text{-C}_5\text{H}_6$ at 873 K and $\varphi = 0.5$.

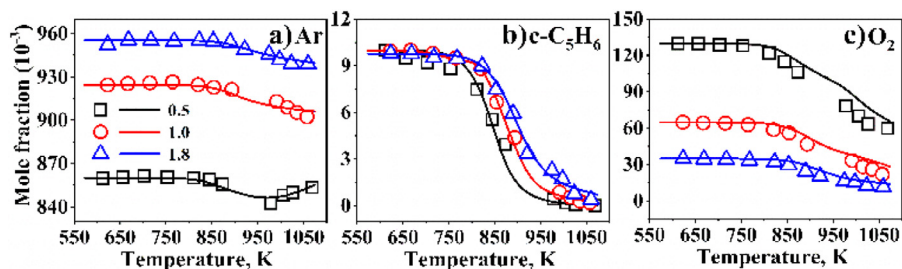


Fig. 2. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) Ar, (b) $c\text{-C}_5\text{H}_6$ and (c) O_2 in the $c\text{-C}_5\text{H}_6$ oxidation. $\varphi = 0.5$: black square and solid line; $\varphi = 1.0$: red circle and solid line; $\varphi = 1.8$: blue triangle and solid line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figs. 2 and 3 show the experimental and simulated mole fraction profiles of major species ($c\text{-C}_5\text{H}_6$, Ar, O_2 , CO, CO_2 , H_2 and H_2O). Generally the current model can reasonably reproduce the consumption of $c\text{-C}_5\text{H}_6$ and O_2 and the formation of four major products. It is noted that Ar is an inert gas in the oxidation, which is very stable and does not participate in any reaction. The change of Ar mole fraction (Fig. 2(a)) shows the variation characteristics of the total molar amount during the $c\text{-C}_5\text{H}_6$ oxidation, which can be attributed to that the increasing of temperature and volume of total gas products lead to gas expansion. Figs. 2(b) and (c) show that with φ increasing, the initial consumption temperature of $c\text{-C}_5\text{H}_6$ and O_2 shifts to higher temperatures and the reaction temperature zone becomes wider. From $\varphi = 0.5$ to 1.8, the initial consumption temperature of $c\text{-C}_5\text{H}_6$ and O_2 changes from 753 K to 823 K and the complete consumption temperature of $c\text{-C}_5\text{H}_6$ changes from 978 K to 1069 K.

Fig. 3(a) shows that the mole fraction of CO increases firstly and then decreases under the lean condition, while it becomes a final product under the stoichiometric and rich conditions. It can be illustrated from the formation pathways of CO: $\text{HCO} + \text{O}_2 \rightarrow \text{CO} + \text{HO}_2$ and $\text{HO}_2 + c\text{-C}_5\text{H}_5 \rightarrow \text{C}_2\text{H}_4 + \text{C}_2\text{H} + \text{CO} + \text{OH}$ and the consumption pathway of CO: $\text{CO} + \text{OH} \rightarrow \text{CO}_2 + \text{H}$ under all equivalence conditions. The decreased OH radical concentration with increasing equivalence ratio leads to the accumulation of CO

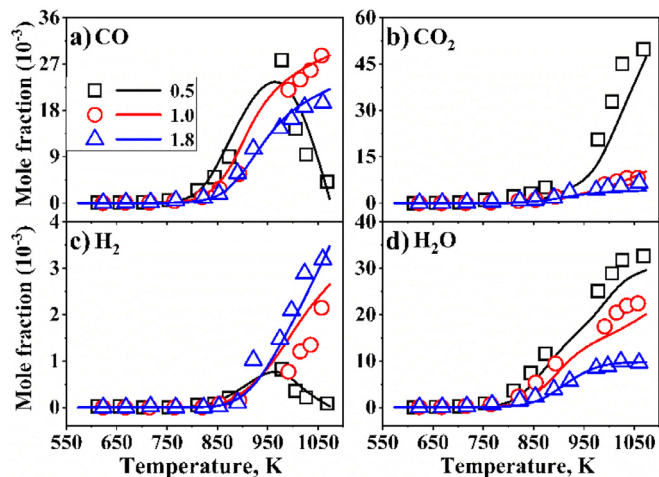


Fig. 3. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) CO, (b) CO_2 , (c) H_2 and (d) H_2O in the $c\text{-C}_5\text{H}_6$ oxidation. $\varphi = 0.5$: black square and solid line; $\varphi = 1.0$: red circle and solid line; $\varphi = 1.8$: blue triangle and solid line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

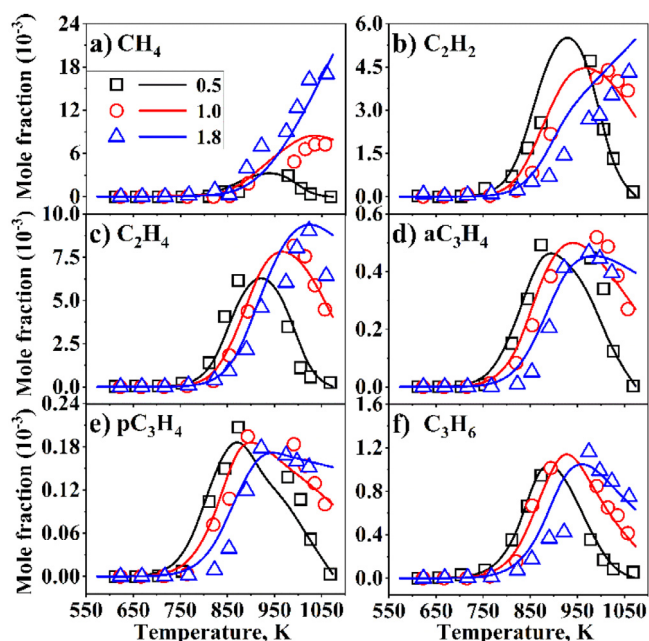


Fig. 4. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) CH_4 , (b) C_2H_2 , (c) C_2H_4 , (d) aC_3H_4 , (e) pC_3H_4 and (f) C_3H_6 in the $\text{c-C}_5\text{H}_6$ oxidation. $\varphi = 0.5$: black square and solid line; $\varphi = 1.0$: red circle and solid line; $\varphi = 1.8$: blue triangle and solid line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

at $\varphi = 1.0$ and 1.8. Fig. 3(b) shows the mole fraction of CO_2 increases monotonically as the increasing of temperatures. The final production of CO_2 decreases with the increasing equivalence ratios, which is due to the formation paths of CO_2 involving more O_2 and CO : $\text{HCCO} + \text{O}_2 \rightarrow \text{CO}_2 + \text{CO} + \text{H}$ and $\text{CO} + \text{OH} \rightarrow \text{CO}_2 + \text{H}$. Fig. 3(c) shows the profile trend of H_2 mole fraction is quite similar to that of CO , which exhibits a peak-shape profile under the lean condition and increases monotonically under stoichiometric and rich conditions. The ROP analysis explains that H_2 is mainly formed from the H abstraction reactions of $\text{c-C}_5\text{H}_6$ and CH_2O by H attacking and consumed via the reaction: $\text{H}_2 + \text{OH} \rightarrow \text{H} + \text{H}_2\text{O}$. With φ increasing from 0.5 to 1.8, the increased H radical concentration and decreased OH radical concentration promote the accumulation of H_2 . Fig. 3(d) shows the mole fraction of H_2O increase dramatically with the increasing of temperature and decreasing of equivalence ratio. It can be attributed to the formation pathways of H_2O through the H-abstraction reactions of $\text{c-C}_5\text{H}_6$ and CH_2O by OH attacking.

4.1.3. Intermediates

Figs. 4–6 present the experimental and simulated results of 18 intermediates as a function of temperature. Fig. 4 depicts the mole fraction profiles of the C1–C3 light hydrocarbons including CH_4 , C_2H_2 , C_2H_4 , aC_3H_4 , pC_3H_4 and C_3H_6 , whose initial formation temperatures and peak value temperatures increase from lean to rich condition. Figs. 4(a) and (b) show that the mole fraction of CH_4 and C_2H_2 exhibit peak-shaped profiles at $\varphi = 0.5$ and 1.0, while it increases continually at $\varphi = 1.8$. The ROP analysis shows that more than 95% of CH_4 is generated from the H-abstraction reaction of $\text{c-C}_5\text{H}_6$ via CH_3 attacking and consumed by the H-abstraction reaction of CH_4 via OH attacking with the contributions of 91% and 74% at $\varphi = 0.5$ and 1.8, respectively. With φ increasing, the reduction of OH concentration causes the accumulation of CH_4 . 86% and 82% of C_2H_2 are produced through the reaction: $\text{c-C}_5\text{H}_6 + \text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2 + \text{HCO}$ and 12% and 11% of C_2H_2 are formed by the reaction: $\text{C}_5\text{H}_3\text{O} + \text{O}_2 \rightarrow \text{C}_2\text{H}_2 + \text{CO}_2 + \text{HCCO}$

at $\varphi = 0.5$ and 1.8, respectively. The elevated equivalence ratio reduces the concentrations of OH and O_2 and causes the decrease of C_2H_2 formation. The C_2H_2 is consumed via the reactions: $\text{C}_2\text{H}_2 + \text{O} \rightarrow \text{HCCO} + \text{H}$ and $\text{C}_2\text{H}_2 + \text{c-C}_5\text{H}_5 \rightarrow \text{c-C}_7\text{H}_7$ and the decreased O content improves the accumulation of C_2H_2 from $\varphi = 0.5$ to 1.8.

Figs. 4(c)–(f) show that the trends of mole fraction profiles of C_2H_4 , aC_3H_4 , pC_3H_4 and C_3H_6 are quite similar under all equivalence conditions, which first increase and then decrease. According to ROP analysis, 72% and 48% of C_2H_4 is formed by the reaction: $\text{c-C}_5\text{H}_6 + \text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2 + \text{HCO}$ and 18% and 29% of C_2H_4 is generated by the reaction: $\text{c-C}_5\text{H}_6 + \text{C}_2\text{H}_3 \rightarrow \text{C}_2\text{H}_4 + \text{c-C}_5\text{H}_5$ at $\varphi = 0.5$ and 1.8, respectively. The consumption paths of C_2H_4 are basically consistent from $\varphi = 0.5$ to 1.8 and mainly include two reactions: $\text{C}_2\text{H}_4 + \text{H} \rightarrow \text{C}_2\text{H}_5$ and $\text{C}_2\text{H}_4 + \text{HO}_2 \rightarrow \text{CH}_2\text{OCH}_2 + \text{OH}$. The contribution of the former reaction is varied from 29% to 40% with the increase of equivalence ratios and that of the latter reaction is varied from 26% to 23%. For aC_3H_4 and pC_3H_4 , the generation and consumption paths are also basically identical. Taking aC_3H_4 as an example to demonstrate, over 90% of aC_3H_4 is yielded through the reaction: $\text{c-C}_5\text{H}_6 + \text{O} \rightarrow \text{CH}_2\text{O} + \text{aC}_3\text{H}_4$. About 53% and 33% of aC_3H_4 are consumed by two reactions: $\text{aC}_3\text{H}_4 + \text{HO}_2 \rightarrow \text{C}_3\text{H}_3 + \text{H}_2\text{O}_2$ and $\text{c-C}_5\text{H}_5 + \text{aC}_3\text{H}_4 \rightarrow \text{A1} + \text{C}_2\text{H}_3$ at $\varphi = 0.5$, respectively. As φ increases from 0.5 to 1.8, the contribution of the former reaction falls to 38% and the latter reaction rises to 39% because of the lower concentration of O_2 . For C_3H_6 , the ROP analysis indicates that 73% of C_3H_6 is formed through two reactions: $\text{aC}_3\text{H}_5 + \text{HO}_2 \rightarrow \text{C}_3\text{H}_6 + \text{O}_2$ and $\text{c-C}_5\text{H}_6 + \text{O}_2 \rightarrow \text{C}_3\text{H}_6 + 2\text{CO}$ at $\varphi = 0.5$, while the contribution of these two reactions is dropped by 6% at $\varphi = 1.8$. The consumption paths of C_3H_6 are mainly via the H-abstraction reactions by OH/H attacking and the combination reaction with H to produce isopropyl radical.

Fig. 5 displays the mole fraction profiles of the oxygenated intermediates including CH_2O , CH_3CHO and $\text{C}_2\text{H}_3\text{CHO}$, whose peaks shift toward higher temperature region as the elevated equivalence ratio. In Fig. 5(a), the ROP analysis reveals that CH_2O is formed mainly through three pathways: $\text{iC}_4\text{H}_5 + \text{O}_2 \rightarrow \text{C}_2\text{H}_3\text{CO} + \text{CH}_2\text{O}$ (43% at $\varphi = 0.5$ and 48% at $\varphi = 1.8$, respectively), $\text{C}_2\text{H}_3 + \text{O}_2 \rightarrow \text{HCO} + \text{CH}_2\text{O}$ (19% at $\varphi = 0.5$ and 1.8, respectively) and propenyl-oxyl ($\text{C}_3\text{H}_5\text{O}$) + OH $\rightarrow \text{C}_2\text{H}_3 + \text{CH}_2\text{O}$ (12% at $\varphi = 0.5$ and 10% at $\varphi = 1.8$, respectively). The consumption of CH_2O is mainly through the H-abstraction reactions by $\text{HO}_2/\text{OH}/\text{O}/\text{H}$ radicals attacking, in which the contribution by H attacking increases with the φ increasing. At $\varphi = 0.5$ and 1.8, the H-abstraction reactions by $\text{HO}_2/\text{OH}/\text{O}$ contribute about 61%/28%/3% and 55%/23%/3% and the H-atom abstraction reaction by H contributes about 8% and 17%, respectively.

In Fig. 5(b), the ROP analysis indicates that 74% and 52% of CH_3CHO are produced by the metathesis of $\text{C}_2\text{H}_3\text{CHO}$ with OH and 24% and 41% of CH_3CHO are formed by the metathesis of vinyl oxy radical (CH_2CHO) with HO_2 , respectively. Similarly to the consumption of CH_2O , CH_3CHO is mainly consumed through the H-abstraction reactions by H/ HO_2 /OH. At $\varphi = 0.5$ and 1.8, the H-abstraction reactions by HO_2/OH contribute about 45%/28% and 34%/20% and that of H radical contributes about 20% and 41%, respectively. In Fig. 5(c), the $\text{C}_2\text{H}_3\text{CHO}$ mainly comes from the combination reactions of $\text{c-C}_5\text{H}_6/\text{nC}_4\text{H}_5/\text{iC}_4\text{H}_5$ with O_2 under both the lean and rich conditions, which strongly depend on the concentration of O_2 . The decreasing of O_2 concentration at the rich mixture limits the rates of these reactions and consequently decreases the concentration of $\text{C}_2\text{H}_3\text{CHO}$. At $\varphi = 0.5$ and 1.8, 60% and 46% of $\text{C}_2\text{H}_3\text{CHO}$ are consumed by the combination reactions with OH: $\text{C}_2\text{H}_3\text{CHO} + \text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{CO}_2 + \text{H}$ and $\text{C}_2\text{H}_3\text{CHO} + \text{OH} \rightarrow \text{CH}_3\text{CHO} + \text{HCO}$, and 12% and 25% of $\text{C}_2\text{H}_3\text{CHO}$ are consumed by the combination reaction with H: $\text{C}_2\text{H}_3\text{CHO} + \text{H} \rightarrow \text{C}_2\text{H}_4 + \text{HCO}$, respectively.

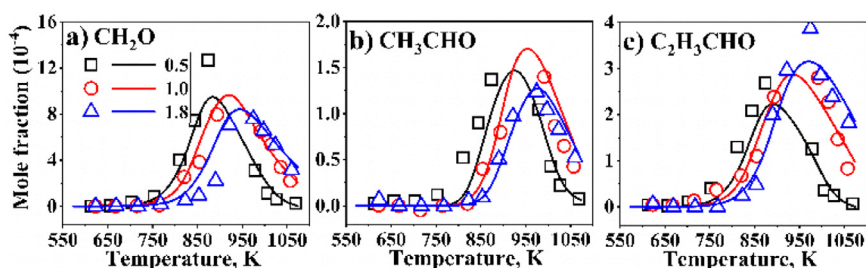


Fig. 5. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) CH_2O , (b) CH_3CHO and (c) $\text{C}_2\text{H}_3\text{CHO}$ in the $c\text{-C}_5\text{H}_6$ oxidation. $\varphi = 0.5$: black square and solid line; $\varphi = 1.0$: red circle and solid line; $\varphi = 1.8$: blue triangle and solid line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

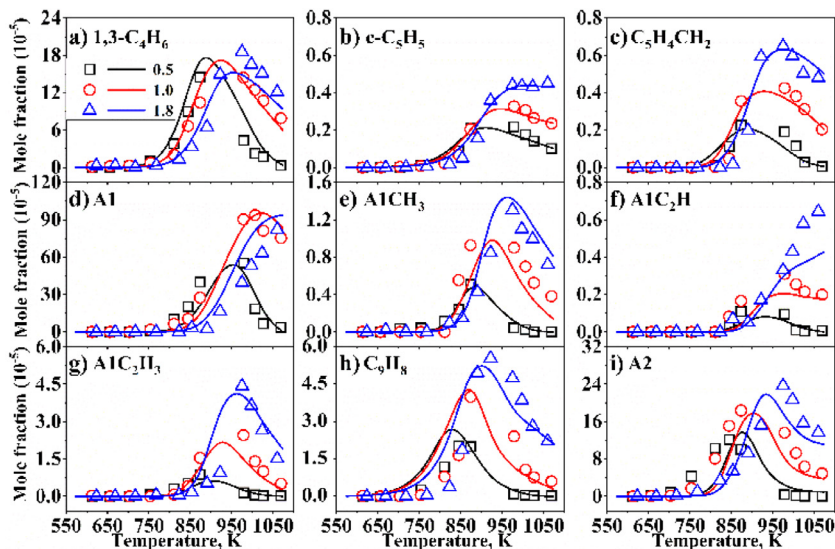


Fig. 6. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) C_4H_6 , (b) $c\text{-C}_5\text{H}_5$, (c) $\text{C}_5\text{H}_4\text{CH}_2$, (d) A1 , (e) A1CH_3 , (f) $\text{A1C}_2\text{H}$, (g) $\text{A1C}_2\text{H}_3$, (h) C_9H_8 and (i) A2 in the $c\text{-C}_5\text{H}_6$ oxidation. $\varphi = 0.5$: black square and solid line; $\varphi = 1.0$: red circle and solid line; $\varphi = 1.8$: blue triangle and solid line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 6 shows the mole fraction profiles of aromatic and related species including C_4H_6 , $c\text{-C}_5\text{H}_5$, $\text{C}_5\text{H}_4\text{CH}_2$, A1 , A1CH_3 , $\text{A1C}_2\text{H}$, $\text{A1C}_2\text{H}_3$, C_9H_8 and A2 , which are crucial in the combustion of $c\text{-C}_5\text{H}_6$ due to the significant role in the PAHs formation. In general, most peak values increase as φ increasing besides those of C_4H_6 . In Fig. 6(a), the ROP analysis indicates that the formation paths of C_4H_6 include the H-abstraction reactions of 1-buten-4-yl radical (C_4H_7) by O_2 and the metathesis of $c\text{-C}_5\text{H}_6$ with OH and O. The contribution of the H-abstraction and the metathesis is approximately equal at $\varphi = 0.5$ and 1.8, which are 10% and 80% to the production of C_4H_6 , respectively. The consumption paths of C_4H_6 are also identical under all conditions containing the recombination of C_4H_6 with H to form 1-buten-3-yl radical (SAXC_4H_7) and C_4H_7 and the combination reaction C_4H_6 with C_3H_3 to produce $\text{A1CH}_3 + \text{H}$. The contribution of latter reaction increases from 20% to 37% with the increasing of equivalence ratio, which causes the increasing trends of speciation profiles of A1CH_3 . In Fig. 6(b), above 85% of $c\text{-C}_5\text{H}_5$ is yielded through the H-abstraction of $c\text{-C}_5\text{H}_6$ by OH, HO_2 , CH_3 and H attacking at all equivalence ratios. The $c\text{-C}_5\text{H}_5$ is mainly consumed through the combination reactions with HO_2 to generate $i\text{C}_4\text{H}_5$ and C_4H_4 and the recombination reaction with itself to produce fulvalene isomers. In Fig. 6(c), more than 96% of $\text{C}_5\text{H}_4\text{CH}_2$ is formed by the H-abstraction reaction of $\text{C}_5\text{H}_4\text{CH}_3$ by O_2 and the combination between C_3H_3 and aC_3H_5 under all equivalence conditions. The $\text{C}_5\text{H}_4\text{CH}_2$ is transformed to A1 through the reaction sequence: $\text{C}_5\text{H}_4\text{CH}_2 \rightarrow \text{C}_5\text{H}_5\text{CH}_2\text{-1/C}_5\text{H}_5\text{CH}_2\text{-2/C}_5\text{H}_5\text{CH}_2\text{-3} \rightarrow \text{A1}$.

In Fig. 6(d), the ROP analysis explains the A1 is produced by the combination reactions of $c\text{-C}_5\text{H}_5 + \text{aC}_3\text{H}_4/\text{pC}_3\text{H}_4$ and $\text{C}_5\text{H}_5\text{CH}_3 + \text{OH}$. The contribution of $\text{C}_5\text{H}_5\text{CH}_3 + \text{OH}$ is markedly reduced by 12% from $\varphi = 0.5$ to 1.8 causing the lower mole fraction of A1 in initial period. With the increasing of temperature, the A1 is consumed through the H-abstraction reaction by OH attacking and the decreased OH concentration reduces the consumption of A1 from the lean to rich condition. As mentioned earlier (Fig. 6(e)), 86% and 77% of A1CH_3 are formed by the combination reaction C_4H_6 with C_3H_3 and extra 5% and 16% of A1CH_3 comes from the combination between $c\text{-C}_5\text{H}_6 + \text{C}_2\text{H}_3$ at $\varphi = 0.5$ and 1.8, respectively. The consumption of A1CH_3 is mainly through the H-abstraction by OH and H to generate A1CH_2 and methylphenyl ($\text{C}_6\text{H}_4\text{CH}_3$), and the ipso-substitution reaction through H to produce A1 and CH_3 . However, the dominant consumption paths are different from lean to rich condition. At $\varphi = 0.5$, the H-abstraction reactions by OH are the major consumption pathways of A1CH_3 with the total contribution of about 68%, while at $\varphi = 1.8$ the H-abstraction reaction and the ipso-substitution reaction by H become more important with the total contribution of approximately 50%. In Fig. 6(f), the ROP analysis shows more than 90% of $\text{A1C}_2\text{H}$ is formed by the H elimination reaction of 1-phenyl-vinyl radical (A1CHCH) under all equivalence conditions, which can be produced by the H-abstraction reactions of $\text{A1C}_2\text{H}_3$. The $\text{A1C}_2\text{H}$ is mainly consumed via the recombination with H to yield 2-phenyl-vinyl radical (A1CCH_2) with a contribution of 62% and 76%

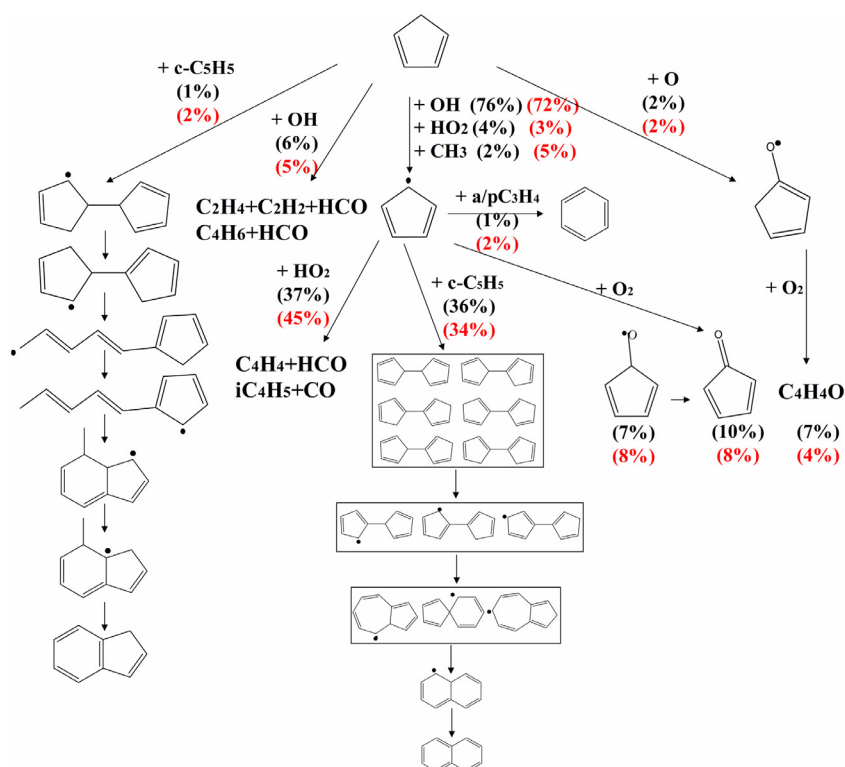


Fig. 7. Reaction networks of $c\text{-C}_5\text{H}_6$ oxidation shown with a conversion of approximately 50% at $\varphi = 0.5$ (848 K): black numbers and $\varphi = 1.8$ (898 K): red numbers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

at $\varphi = 0.5$ and 1.8, respectively. The contribution of combination reactions of $\text{A1C}_2\text{H}$ with OH/O to produce A1/A1- is reduced by 10% from $\varphi = 0.5$ to 1.8.

In Fig. 6(g), the $\text{A1C}_2\text{H}_3$ is mainly produced by the recombination (60%) of $c\text{-C}_5\text{H}_5$ with C_3H_3 and the H-atom abstraction reactions (35%) of 2-phenylethyl ($\text{A1CH}_2\text{CH}_2$) and 1-phenylethyl (A1CHCH_3) by O_2 . About 86% of $\text{A1C}_2\text{H}_3$ is consumed through the recombination reactions with H to generate $\text{A1CH}_2\text{CH}_2$ and A1CHCH_3 under all equivalence conditions. In Fig. 6(h), the ROP analysis indicates that most of C_9H_8 is from the combination between $c\text{-C}_5\text{H}_6$ and $c\text{-C}_5\text{H}_5$. The C_9H_8 is consumed through the H-abstraction by HO_2 , OH , O and H radicals to form C_9H_7 and the combination reaction with O to produce indenone ($\text{C}_9\text{H}_6\text{O}$), which the contribution by H attacking increases with the φ increasing. At $\varphi = 0.5$ and 1.8, the H-abstraction reaction by H contributes 25% and 46% to the consumption of C_9H_8 and the combination reaction with HO_2 contributes 47% and 36%, respectively. In Fig. 6(i), almost all of A2 is formed by the fulvalanyl radical isomers via losing a H atom, which are initiated by the self-recombination of $c\text{-C}_5\text{H}_5$. The A2 is mainly consumed through the H-abstraction reaction by OH attacking to form naphthyl (A2-) and the decreased OH concentration as φ increasing causes the improvement of mole fraction of A2 at the higher temperature region. Additionally, the variations of the mole fraction of aromatics as the equivalence ratios are quite consistent with the oxidation experimental results of $c\text{-C}_5\text{H}_6$ in the flow reactor of Butler et al. [23].

4.2. Rate of production and sensitivity analysis

In order to provide insights into the oxidation chemistry of $c\text{-C}_5\text{H}_6$, the reaction flux analysis has been performed for equivalence ratios of $\varphi = 0.5$ (848 K) and $\varphi = 1.8$ (898 K) corresponding to around 50% conversion of $c\text{-C}_5\text{H}_6$. As shown in Fig. 7, the main consumption channels of $c\text{-C}_5\text{H}_6$ under the lean and rich conditions are extremely similar although existing some subtle

differences. At both equivalence ratios, the ROP analysis indicates that $c\text{-C}_5\text{H}_6$ is mainly consumed through the H-abstraction reactions to generate $c\text{-C}_5\text{H}_5$ by OH , HO_2 and CH_3 , which the OH attacking is the most dominant consumption pathway for both cases. At $\varphi = 0.5$, about 76% of $c\text{-C}_5\text{H}_6$ is consumed through the H-abstraction reaction by OH , whereas the contribution of that decreases to 72% at $\varphi = 1.8$. Instead, the total contribution of H-abstraction reaction by CH_3 to the consumption of $c\text{-C}_5\text{H}_6$ rises to 5% under the rich condition. In addition, the oxidation reactions of $c\text{-C}_5\text{H}_6$ by OH to generate $\text{C}_2\text{H}_2 + \text{C}_2\text{H}_4$ and C_4H_6 are also significant consumption paths of $c\text{-C}_5\text{H}_6$, which contribute around 6% and 5% at $\varphi = 0.5$ and 1.8, respectively. A small amount of $c\text{-C}_5\text{H}_6$ consumption contributes to the formation of $\text{C}_5\text{H}_5\text{O13}$ and C_9H_8 , whose total contribution is slightly increased from 3% to 4% as the increasing of equivalence ratios. Subsequently, the $c\text{-C}_5\text{H}_5$ is mainly consumed through the reactions with HO_2 to generate C_4H_4 and iC_4H_5 , which contribute 22% and 15% at the lean condition, respectively. The 36% of $c\text{-C}_5\text{H}_5$ is consumed by the recombination to form A2 , which slightly decreases to 34% at the rich condition. Since the formation pathways of C_9H_8 and A2 from $c\text{-C}_5\text{H}_6$ and $c\text{-C}_5\text{H}_5$ have been described in detail in Section 3.2.2, it will not be discussed here. The recombination of $c\text{-C}_5\text{H}_5$ with $\text{aC}_3\text{H}_4/\text{pC}_3\text{H}_4$ to eventually produce A1 contributes approximately 1% and 2% to the consumption of $c\text{-C}_5\text{H}_5$ at $\varphi = 0.5$ and 1.8, respectively. Additionally, 7%, 10% and 7% of $c\text{-C}_5\text{H}_5$ are consumed by the combination with O_2 to form $\text{C}_5\text{H}_5\text{O24}$, $\text{C}_5\text{H}_4\text{O}$ and vinyl ketene ($\text{C}_4\text{H}_4\text{O}$) at $\varphi = 0.5$, respectively. The $\text{C}_5\text{H}_5\text{O24}$ is further consumed to produce $\text{C}_5\text{H}_4\text{O}$. The subsequent decomposition of $\text{C}_5\text{H}_4\text{O}$ is quite similar at both equivalence ratios and can produce C_4H_4 , which is the main source of aC_3H_5 .

To distinguish the key reactions for the consumption of 1,3-cyclopentadiene, a local sensitivity analysis was conducted using the current model at $\varphi = 0.5$ (848 K), 1.0 (873 K) and 1.8 (898 K) corresponding to the conversion of about 50%, shown in Fig. 8. The sensitivity analysis shows that the reactions between $c\text{-C}_5\text{H}_5$ and

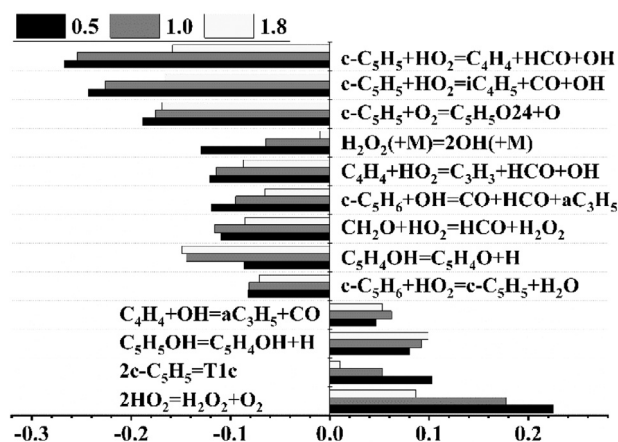


Fig. 8. Sensitivity analysis of $c\text{-C}_5\text{H}_6$ oxidation at $\varphi = 0.5$ (848 K), 1.0 (873 K) and 1.8 (898 K) corresponding to the conversion of about 50%.

HO_2 are the dominant promoting reactions for $c\text{-C}_5\text{H}_6$ oxidation at $\varphi = 0.5$ and 1.0. However, the reactions of $c\text{-C}_5\text{H}_5$ and O_2 is the most promoting reaction for oxidation of $c\text{-C}_5\text{H}_6$ at $\varphi = 1.8$. The reaction of $2\text{HO}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$ is the most inhibiting elementary step in the consumption of $c\text{-C}_5\text{H}_6$ at $\varphi = 0.5$ and 1.0. Similar to the promoting influence at 1.8, the reaction of $\text{C}_5\text{H}_5\text{OH} \rightarrow \text{C}_5\text{H}_4\text{OH} + \text{H}$ has the most negative influence on the consumption of $c\text{-C}_5\text{H}_6$. The ROP analysis explains that the extensive HO_2 is yielded from the oxidation reaction of HCO under the all conditions and then converts into OH through the combination reactions of $c\text{-C}_5\text{H}_5 + \text{HO}_2$ and $\text{C}_4\text{H}_4 + \text{HO}_2$. Therefore, these reactions exhibit the positive effects, whereas the consumption of HO_2 to form $\text{H}_2\text{O}_2 + \text{O}_2$ has the negative effect on the consumption of $c\text{-C}_5\text{H}_6$.

4.3. Comparison with previous 1,3-cyclopentadiene models

The detailed kinetic models involving the $c\text{-C}_5\text{H}_6$ oxidation of Yuan et al. [13], Djokic et al. [16] (referred as the Polimi_1311 model) and Zhou et al. [60] (referred as the Aramco 3.0 model) were used to simulate the $c\text{-C}_5\text{H}_6$ oxidation at $\varphi = 0.5$ in compari-

son with the current model. The comparisons of experimental and simulated results using the four models are displayed in Figs. 9 and 10. In Fig. 9, both the current model and the Yuan model can accurately capture the consumption of $c\text{-C}_5\text{H}_6$ and O_2 , while the Aramco model under-predicts and the Polimi_1311 model seriously over-predicts the reactivity of $c\text{-C}_5\text{H}_6$. The sensitivity analysis shows that main reason for the difference between the Polimi_1311 model and other models is the divergences of small molecular mechanism, shown in Fig. S5 of Supplementary data 1. By replacing the major sensitivity reactions in the Polimi_1311 model with those of current model, the consumption rate of $c\text{-C}_5\text{H}_6$ slows down significantly using the revised Polimi_1311 model (Fig. S6 of Supplementary data 1).

Except the Polimi_1311 model, the other three models can well simulate the formation of CO , CO_2 , H_2O and C_2H_2 within the limits of experimental errors. The Aramco model also performs well in predicting the formation of CH_4 and C_2H_4 , while overestimates the generation of H_2 . The Yuan model over-predicts the formation of H_2 , CH_4 and C_2H_4 . Due to the overestimated initial consumption of $c\text{-C}_5\text{H}_6$, the Djokic model cannot well capture the formation temperature and maximum mole fraction of products. In Fig. 10, the simulated results of C_3H_6 , $\text{C}_2\text{H}_3\text{CHO}$ and A1 accord well with experimental data using the current model and the Aramco 3.0 model, while the Aramco 3.0 model underestimates the formation of CH_2O , CH_3CHO and C_4H_6 . It should be noted that there is no aromatics sub-mechanisms larger than benzene in the Aramco 3.0 model, therefore these species were not compared. The Yuan model can capture effectively the trends of these products in Fig. 10, while under-predicts the formation of C_3H_6 , CH_2O , A1, C_9H_8 and A2 and over-predicts the generation of C_4H_6 and A1CH_3 . In summary, it is observed that the current model performs better in reproducing the experimental results than other three models.

4.4. Model validation with other literature data

Besides the JSR oxidation data reported in the present work, a large number of experimental data available in literature were also adopted to validate the current model, including three kinds of flow reactors pyrolysis and oxidation of the $c\text{-C}_5\text{H}_6$ under the atmospheric pressure, listed in Table S4 of Supplementary data 1.

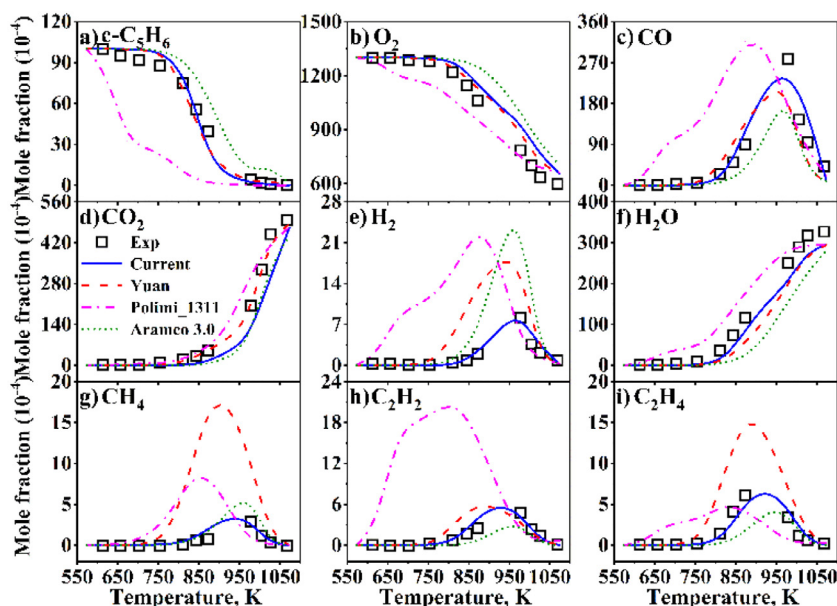


Fig. 9. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) $c\text{-C}_5\text{H}_6$, (b) O_2 , (c) CO , (d) CO_2 , (e) H_2 , (f) H_2O , (g) CH_4 , (h) C_2H_2 and (i) C_2H_4 in the $c\text{-C}_5\text{H}_6$ oxidation at $\varphi = 0.5$ using the current model, Yuan model [13], Polimi_1311 model [16] and Aramco 3.0 model [60].

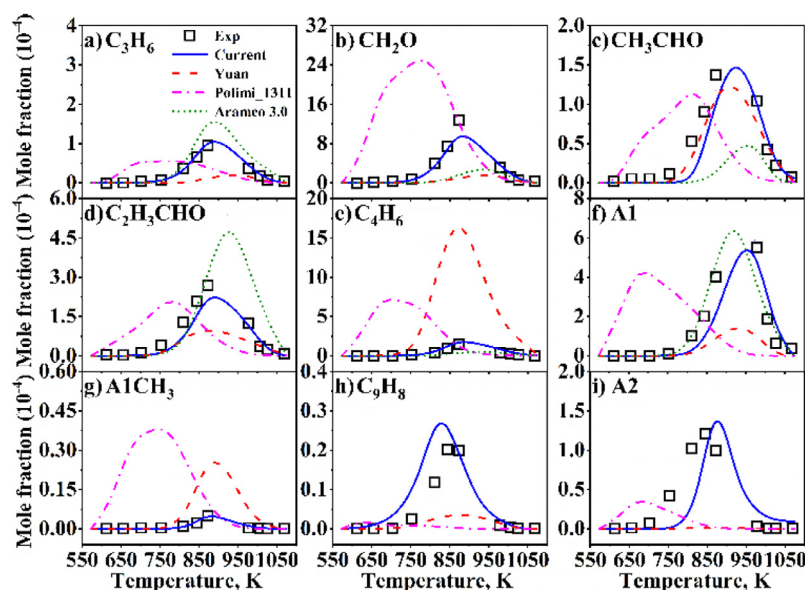


Fig. 10. Experimental (symbols) and simulated (lines) mole fraction profiles of (a) C_3H_6 , (b) CH_2O , (c) CH_3CHO , (d) C_2H_3CHO , (e) C_4H_6 , (f) A1, (g) $A1CH_3$, (h) C_9H_8 and (i) A2 in the $c-C_5H_6$ oxidation at $\varphi = 0.5$ using the current model, Yuan model [13], Polimi_1311 model [16] and Aramco 3.0 model [60].

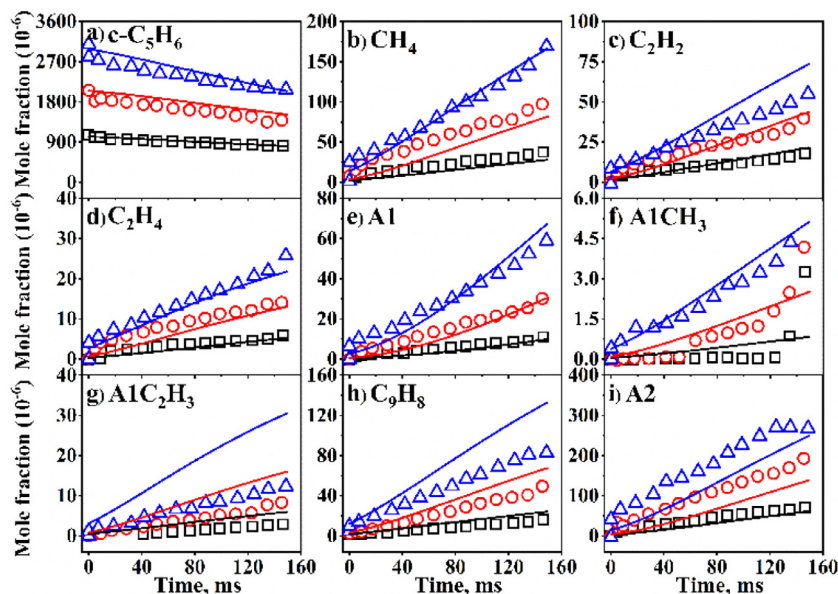


Fig. 11. The effect of varied $c-C_5H_6$ concentration on mole fractions of pyrolysis products with the fixed initial temperature of 1147 K: (a) $c-C_5H_6$, (b) CH_4 , (c) C_2H_2 , (d) C_2H_4 , (e) A1, (f) $A1CH_3$, (g) $A1C_2H_3$, (h) C_9H_8 , and (i) A2. (triangles: $[c-C_5H_6]_0 = 3081$ ppmv, circles: $[c-C_5H_6]_0 = 2083$ ppmv and squares: $[c-C_5H_6]_0 = 1044$ ppmv; Lines: model predictions). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.4.1. Pyrolysis and oxidation in Princeton's flow reactor

Butler and Glassman investigated the atmospheric pyrolysis and oxidation of $c-C_5H_6$ in the Princeton's adiabatic continuous flow reactor (L: 1.0 m and I.D.: 10.0 cm) using a water cooled sample probe and GC-FID analysis [23,24]. The experiments used nitrogen as carrier gas with varied initial $c-C_5H_6$ concentrations from 1000 to 3000 ppm by volume (ppmv), equivalence ratios from fuel lean ($\varphi = 0.6$) to pyrolytic conditions ($\varphi = 100$) and initial temperatures from 1100 to 1200 K. The detailed experimental conditions were shown in Table S5 of Supplementary data 1. These experimental data were used to validate the current model using the Plug Flow Reactor model in the Chemkin-Pro software [87].

Figs. 11 and 12 illustrate experimental and simulated mole fraction profiles of $c-C_5H_6$ and its pyrolysis products with varied initial concentrations and temperatures. The current model can reason-

ably reproduce the consumption of $c-C_5H_6$ and formation of major products, including CH_4 , C_2H_2 , C_2H_4 , A1, $A1CH_3$, $A1C_2H_3$, C_9H_8 , and A2, both in relation to the effects of $c-C_5H_6$ initial concentrations and initial pyrolysis temperatures. It should be noted that the simulated mole fraction are shifted by 20–30 ms in order to be consistent with the fuel conversion rate through adopting the conventional practice used for comparison with the Princeton flow reactor data in literature [13,16,64,95]. Similarly, Fig. 13 presents the comparisons between the experimental and predicted mole fraction profiles of $c-C_5H_6$ and oxidation products under the lean, stoichiometric and rich conditions (0.6–1.6). The model predictions agree well with the experimental data of major species.

Under the pyrolysis condition of Case 1 (Table S5), the decomposition of $c-C_5H_6$ is mainly through the addition reaction with H at the 1 position to produce IC_5H_7 (25%), bond dissociation

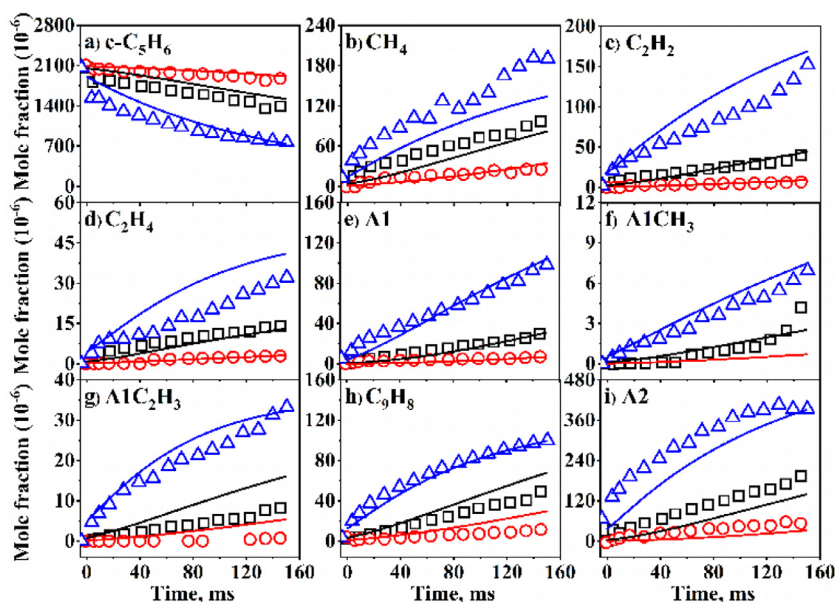


Fig. 12. The effect of varied initial pyrolysis temperature on mole fractions of pyrolysis products with the fixed approximately initial $c\text{-C}_5\text{H}_6$ concentration of 2080 ppmv: (a) $c\text{-C}_5\text{H}_6$, (b) CH_4 , (c) C_2H_2 , (d) C_2H_4 , (e) A1, (f) A1CH_3 , (g) $\text{A1C}_2\text{H}_3$, (h) C_9H_8 , and (i) A2. (triangles: $T_0 = 1202\text{ K}$, squares: $T_0 = 1148\text{ K}$ and circles: $T_0 = 1106\text{ K}$; Lines: model predictions). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

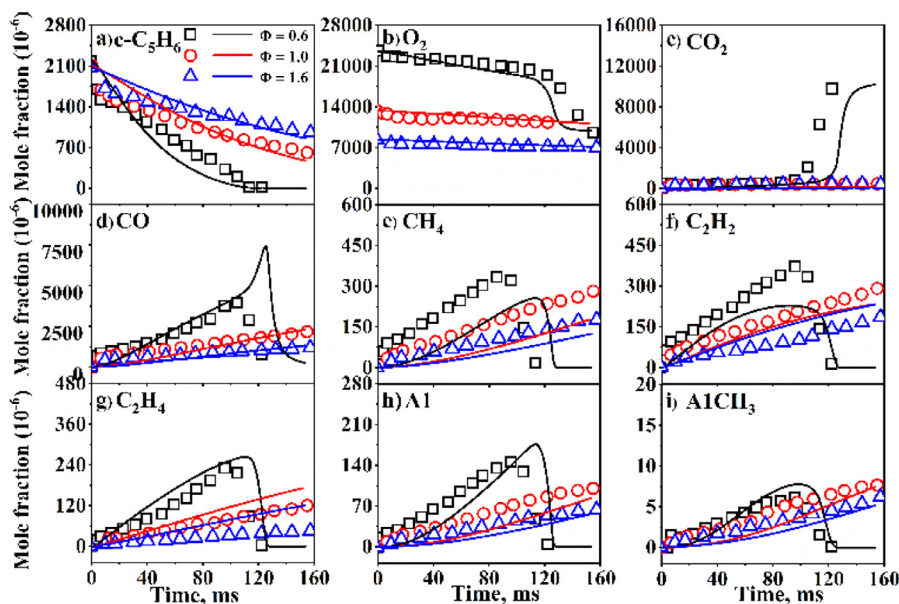


Fig. 13. The mole fractions and simulations of the $c\text{-C}_5\text{H}_6$ and oxidation products at the initial temperature of about 1150 K and initial concentration of approximately 2100 ppmv: (a) $c\text{-C}_5\text{H}_6$, (b) O_2 , (c) CO_2 , (d) CO , (e) CH_4 , (f) C_2H_2 , (g) C_2H_4 , (h) A1 and (i) A1CH_3 .

reactions to form $c\text{-C}_5\text{H}_5 + \text{H}$ (21%), H-atom abstraction reactions by H, CH_3 and C_2H_3 (26%) and the addition reaction with H to produce $\text{aC}_3\text{H}_5 + \text{C}_2\text{H}_2$ (20%). The all of IC_5H_7 is directly converted to $\text{aC}_3\text{H}_5 + \text{C}_2\text{H}_2$. Additionally, about 5% of $c\text{-C}_5\text{H}_6$ is consumed by the recombination with $c\text{-C}_5\text{H}_5$ to produce aromatics, including A1 and C_9H_8 . The ROP analysis shows that the varied the initial concentration of $c\text{-C}_5\text{H}_6$ and initial temperature have only little influence on the major consumption pathways during the $c\text{-C}_5\text{H}_6$ pyrolysis. Under the oxidation condition at $\phi = 1.0$ of Case 7 (Table S5), the H-abstraction reactions by OH, CH_3 , O and HO_2 account for around 82% the consumption of $c\text{-C}_5\text{H}_6$ to form $c\text{-C}_5\text{H}_5$. The rest of $c\text{-C}_5\text{H}_6$ is consumed by the combination with O, OH and O_2 to produce small molecular compounds, including CO, C_2H_2 , C_2H_4 , aC_3H_4 , pC_3H_4 and C_4H_6 . It can be noted that the major consump-

tion paths of $c\text{-C}_5\text{H}_6$ oxidation in the flow reactor are quite similar to those in the JSR oxidation in the high temperature zone, thus the subsequent reactions will not be repeated herein.

4.2.2. Pyrolysis in UGent bench-scale tubular reactor

Djokic et al. studied the $c\text{-C}_5\text{H}_6$ pyrolysis under both low (16.7% $c\text{-C}_5\text{H}_6$ in N_2) and high (4% $c\text{-C}_5\text{H}_6$ in N_2) dilutions [16]. The experiments were conducted in a tubular continuous flow reactor (L: 1.48 m and I.D.: 6.0 mm) under 0.17 MPa and 873–1123 K with the on-line detection of GC $\times$ GC-FID/(TOF-MS). Meanwhile, they refined an existing detailed kinetic model for applications in the low and high dilutions of $c\text{-C}_5\text{H}_6$ pyrolysis, which was validated against not only their experimental results but also previous pyrolysis data and premixed laminar flame speed measurements. Fig. 14 shows

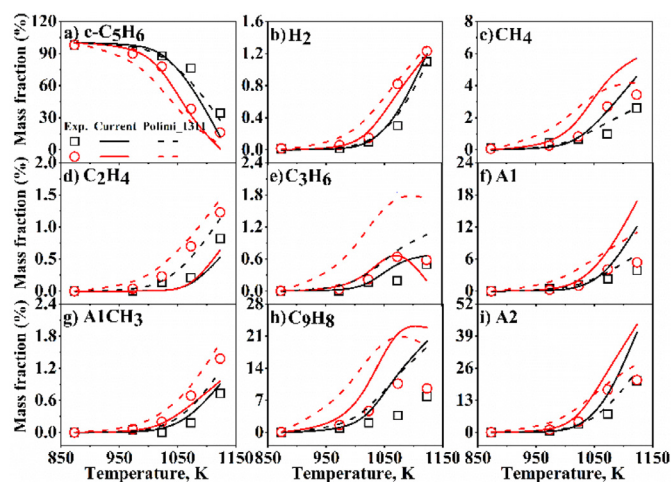


Fig. 14. Experimental (Symbols) [16] and simulated mole fraction profiles (lines) of (a) $c\text{-C}_5\text{H}_6$, (b) H_2 , (c) CH_4 , (d) C_2H_4 , (e) C_3H_6 , (f) A1 , (g) A1CH_3 , (h) C_9H_8 and (i) A2 during the $c\text{-C}_5\text{H}_6$ pyrolysis in a flow reactor. (Experiment: black box: high dilution and red circle: low dilution; Simulation: Current model: solid line and Polimi_1311 model: dotted line [16]). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the comparison of experimental results and simulated mole fraction profiles by the current model and model proposed by Djokic et al. [16]. It can be seen that the current model properly predicts the experimental results at both high and low dilutions. Compared to the simulations of Polimi_1311 model, the current model has an improvement in reproducing the consumption of $c\text{-C}_5\text{H}_6$ and formation of C_3H_6 , though it slightly under-predicts the formation of C_2H_4 .

As displayed in Fig. 14, the initial decomposition temperature of $c\text{-C}_5\text{H}_6$ shifts to the low-temperature region with the decreasing of

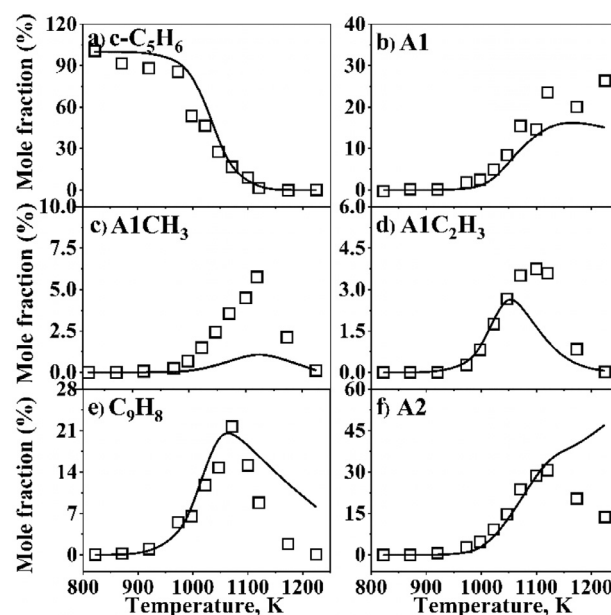


Fig. 15. Experimental (Symbols) [22] and simulated mole fraction profiles (lines) of (a) $c\text{-C}_5\text{H}_6$, (b) A1 , (c) A1CH_3 , (d) $\text{A1C}_2\text{H}_3$, (e) C_9H_8 and (f) A2 during the $c\text{-C}_5\text{H}_6$ pyrolysis in a flow reactor using the current model.

dilution and the A1 , C_9H_8 and A2 are the major products at both dilutions. Taking the low dilution at 1073 K for example, the ROP analysis illustrates that the main decomposition pathways of $c\text{-C}_5\text{H}_6$ are mainly controlled by the H-abstraction reactions (49%) by H , CH_3 , C_2H_3 and aC_3H_5 and the combination reactions (31%) with $c\text{-C}_5\text{H}_5$ to form A1 and $\text{C}_{10}\text{H}_{11}$, in which the $\text{C}_{10}\text{H}_{11}$ is the main source of C_9H_8 . It should be noted that the consumption paths of $c\text{-C}_5\text{H}_6$ are slightly different from these of the Princeton flow

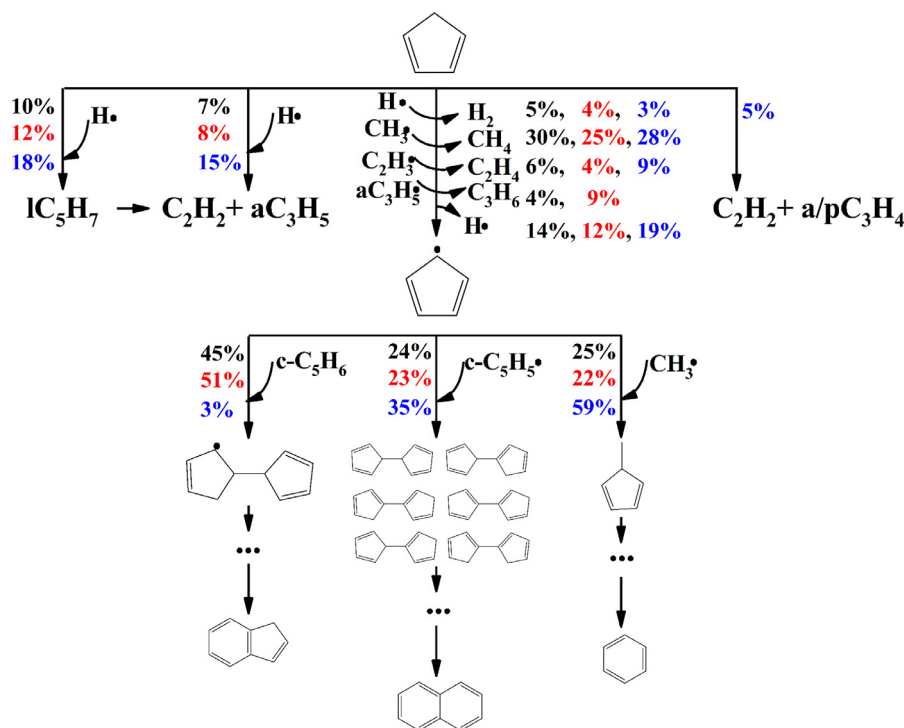


Fig. 16. The reaction pathways of $c\text{-C}_5\text{H}_6$ pyrolysis with approximately 40% conversion at different reactors. Black number: Georgia Tech flow reactor (1023 K, 0.17 MPa); Red number: Ugent flow reactor (1073 K, 0.1 MPa, high dilution); Blue number: Princeton flow reactor (1202 K, 0.1 MPa, 2077 ppmv). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

reactor, which may be contributed to the longer residence time in the UGent flow reactor to promote the contributions of H-abstractions and combination with $\text{c-C}_5\text{H}_5$.

Vervust et al. [38] investigated the co-pyrolysis of $\text{c-C}_5\text{H}_6$ (8.3%) and C_2H_2 (8.3%) in N_2 using the same experimental device as Djokic et al. [16] under 0.17 MPa and 873–1163 K. They presented a detailed kinetic model for the $\text{c-C}_5\text{H}_6$ pyrolysis using an automated reaction network generator (RMG). Figure S7 of Supplementary data 1 shows the comparison of experimental results and simulated mole fraction profiles by the current model and model proposed by Vervust et al. [38]. It can be seen that the current model well predicts the experimental results. Compared to simulations by Vervust model, the current model has a slight improvement in reproducing the formation of A2, though it slightly under-predicts the formation of H_2 and A1CH_3 .

4.3.3. Pyrolysis in Georgia Tech flow tubular reactor

Kim et al. [22] conducted the $\text{c-C}_5\text{H}_6$ pyrolysis experiments in a laminar and continuous flow quartz reactor with 48 cm long and 1.7 cm in diameter under atmospheric pressure and 823–1223 K at a nominal residence time of around 3.0 s. During the experimental process, $\text{c-C}_5\text{H}_6$ was diluted to 0.7% molar by N_2 and the gas temperature was nearly uniform in the reaction zone. Due to the long residence time, the major products detected were aromatics including A1, C_9H_8 and A2, which were mainly formed by the recombination and addition reactions of $\text{c-C}_5\text{H}_6$ and $\text{c-C}_5\text{H}_5$. The comparisons of experimental and predicted mole fraction profiles of pyrolysis products are displayed in Fig. 15. In general, the current model can reasonably reproduce the decomposition of 1,3-cyclopentadiene and the generation of pyrolysis products. The prediction of A1CH_3 formation is lower than the measurements, which may be due to the lower rate constants of $\text{A1CH}_2 + \text{H}$. When the temperature is higher than 1173 K, the simulated mole fraction of A2 is still increasing, which is contrary to the experimental measurements. The deviation may be attributed to the lacking of sub-mechanisms of successive growth of larger aromatics and soot formation.

Fig. 16 displays the comparison of the reaction pathways with about 40% $\text{c-C}_5\text{H}_6$ conversion at different reactors, including Georgia Tech flow reactor (1023 K, 0.17 MPa), Ugent flow reactor (1073 K, 0.1 MPa, high dilution) and Princeton flow reactor (1202 K, 0.1 MPa, 2077 ppmv). It can be observed that $\text{c-C}_5\text{H}_6$ is mainly consumed through the H abstraction reactions and C–H bond dissociation reaction to form $\text{c-C}_5\text{H}_5$ with approximately equal proportions in three reactors (59%, 54% and 59%, respectively). However, the contribution of small species generation obviously increases to the consumption of $\text{c-C}_5\text{H}_6$ with the increasing of temperature in three reactors, which are 17%, 20% and 38%, respectively. At the same time, it is also found that the lower pyrolysis temperatures in the Georgia Tech and Ugent flow reactors can form more C_9H_8 and the higher temperature in Princeton flow reactor promotes the formation of A1 and A2. These results are in good agreement with the experimental findings of Kim et al. [22].

5. Conclusions

The jet stirred reactor oxidation of 1,3-cyclopentadiene was studied at the pressure of 0.1 MPa, temperatures from 623 to 1073 K and equivalence ratios of 0.5, 1.0 and 1.8 using SVUV-PIMS. The mole fraction profiles vs. reactor temperature of 25 oxidation species were obtained, including light hydrocarbons, oxygenated and aromatic species with uncertainties of C, H, and O molar balances about 24%, 20% and 14%, respectively. A detailed kinetic model with 245 species and 1638 reactions was constructed based on the theoretical calculation and reported models in literature and validated against the new JSR oxidation data in this work. By the

ROP analysis, the relative reaction networks were provided for the further investigation 1,3-cyclopentadiene oxidation chemistry. The 1,3-cyclopentadiene is mainly consumed through H-atom abstraction reactions by OH, HO_2 and CH_3 to form cyclopentadienyl radical, which is consumed by combination reactions with HO_2 and O_2 to generate small species and recombination reaction to form A2. The dominant consumption pathways of 1,3-cyclopentadiene and cyclopentadienyl radical demonstrate slight differences at different equivalence ratios. For 1,3-cyclopentadiene oxidation from the lean to rich condition, the roles of CH_3 become more important although the dominant reaction is still the H-atom abstraction reaction by OH.

Furthermore, the current model was also validated with literature experimental results of 1,3-cyclopentadiene combustion, including the species concentration profiles in three kinds of flow reactors pyrolysis and oxidation under atmospheric pressure. The performance of the current model in reproducing literature data is reasonably good over a wide range of conditions. This work is helpful to comprehensively understand the pyrolysis and oxidation of 1,3-cyclopentadiene and further investigate the combustion chemistry of aromatics, biomass and high-energy density fuels producing 1,3-cyclopentadiene under the wide experimental conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.10.032.

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